Privacy and data governance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which reader can link which records and what proves the restriction.

Teach purpose, pseudonymity and honest control/retention evidence.

Keep one habit: Model coded-export risk → trace restricted key → distinguish proposed and implemented controls → assess subgroup policy → retain disclosure limits.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of privacy and **data governance**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of research ethics and consent

Reference example, not the only acceptable wording: Teacher explanation: Research ethics considers purpose, burden, voluntary participation and consequences.

Approval to teach does not automatically authorise a research data collection. Use fictional records for this course.

For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in consent and withdrawal handling before collection. Do not implement real recruitment in this exercise.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sensitivity analysis

Reference example, not the only acceptable wording: Teacher explanation: Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices.

It should not secretly select the most favourable result. Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range.

Settings affecting data roles require special caution.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of statistical tests and assumptions

Reference example, not the only acceptable wording: Teacher explanation: A statistical test compares observed **evidence** with a specified null model. A p-value is not the probability that the null hypothesis is true.

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed.

Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Pictured end goal

Explain privacy and **data governance** using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Research ethics and consent

Sensitivity analysis

Statistical tests and assumptions

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Data governance** specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable. Use the supplied constructed scenarios to make the **mechanism** inspectable.

Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of privacy and **data governance**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach governance as an evidence-backed data-flow design

> Assess purpose, readers and remaining linkage routes rather than a name-removal checklist.

Definitions and complete fictional contract

Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Governance item	Complete fictional exercise contract
Named purpose	Task-score analysis using fictional records
Analysis fields needed in the picture	Random participant code and task score
Unnecessary main-export details	Exact birth date, classroom and rare condition are not needed for this toy purpose
Separate key	Code-to-identity link; restricted withdrawal role only
Analyst role	Reads analysis values, not the identity key under this design
Public output in the proposed repair	No row-level public output; any later aggregate needs its own disclosure check

Governance item	Complete fictional exercise contract
Retention in the proposed repair	Demo plan: delete analysis and key at exercise end; this is a proposal, not executed deletion
Subgroup policy for the independent case	Do not report groups smaller than five
Evidence limit	Metadata checks do not prove anonymity, actual access control, consent or safe export

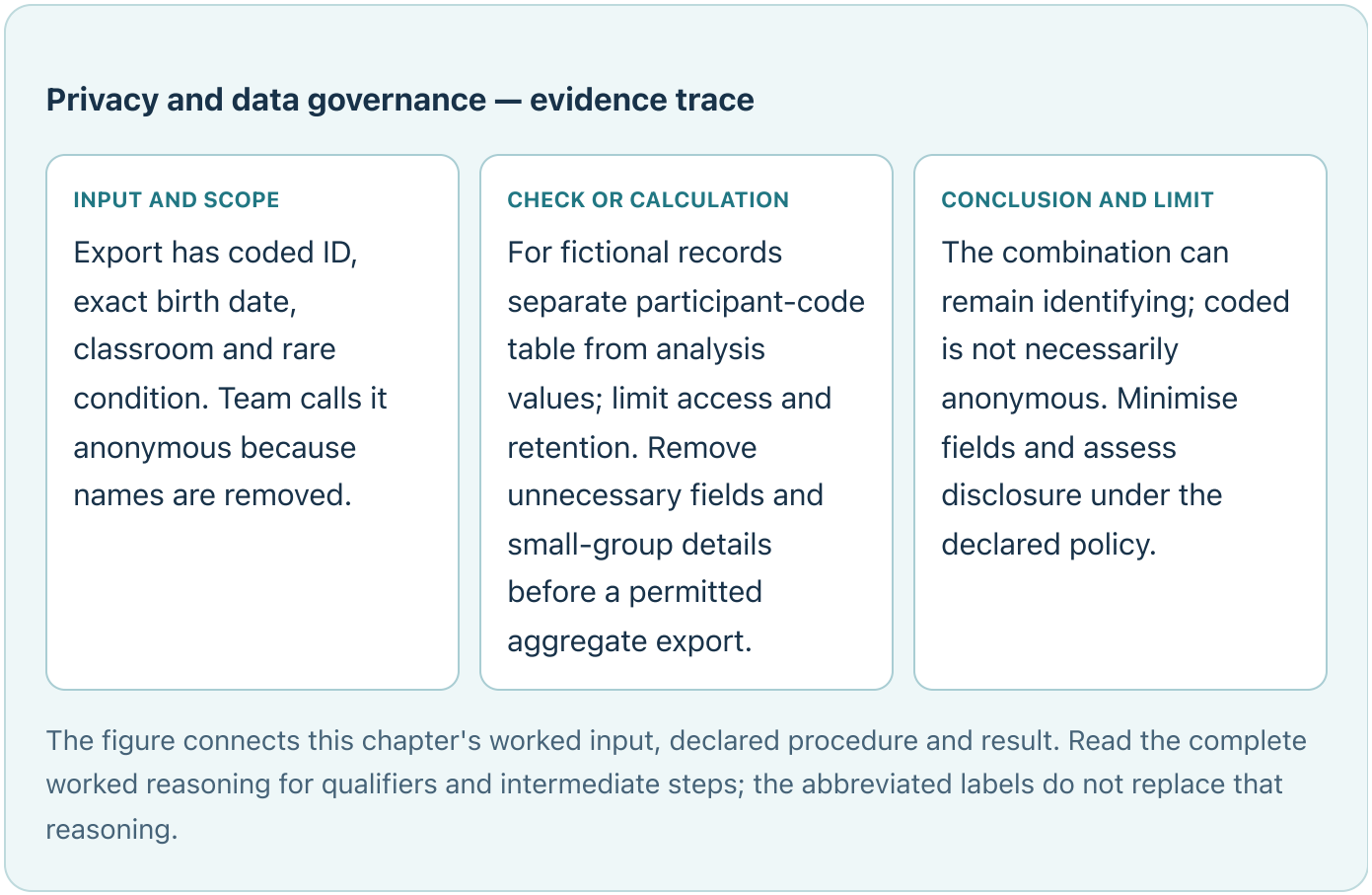
Main field ledger and checked teacher reasoning

Main export field	Reason for concern	Toy-purpose action
Coded ID	May link to other records or a key	Restrict under the declared role plan
Exact birth date	Detailed identifying combination	Remove from the task-score analysis proposal
Classroom	Narrows the possible people	Remove unnecessary detail
Rare condition	Can identify through unusual combinations	Remove from this toy-purpose proposal
No names in export	Does not remove all linkage routes	Do not label it guaranteed anonymous
Missing readers/retention evidence	Plan alone does not prove controls	Record unresolved checks; do not claim actual protection

Main coded ID, birth date, classroom and rare condition can remain identifying in combination; removing names does not establish anonymity. For the explicit toy task-score purpose, unnecessary detail is removed in a proposed plan, with roles and retention recorded. Picture code-to-score and restricted code-to-identity tables remain linkable; publishing the key breaks separation.

Metadata checks do not enforce access or delete records. Independent subgroup of one fails the stated minimum-five policy. Do not publish that subgroup; any suppression or grouping proposal must avoid revealing the hidden value through totals and must stay within the allowed purpose. Meeting the size rule at five alone does not guarantee privacy or permission.

No real export, identity lookup or institutional/legal determination occurs.



Guided: The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Qualification: A size threshold or coding scheme alone does not guarantee privacy.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.

The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Control-evidence and subgroup probes

Governance probe	Checked interpretation
Names removed	Other fields may still identify people
Random codes used	Key or external information can allow linkage
Key is in another file	File separation does not prove reader restrictions
Metadata audit returns no listed issues	Not actual permission, anonymity or disclosure clearance
Retention action written down	Not proof of deletion
One-member subgroup under minimum-five policy	Do not publish the subgroup
Five-member subgroup	Meets only the stated size rule; other risks remain
Merge or suppress categories	Check permitted purpose and whether totals reveal the suppressed cell
Fictional example is well designed	Does not authorize a real export

Separate key design and native diagram

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

The analysis table holds codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling.

The separate picture keeps participant codes and task scores in the analysis table. A restricted key maps those codes to identities for the stated withdrawal purpose. That link is why the records are pseudonymous rather than guaranteed anonymous. The separation reduces routine exposure only if readers actually have the declared restrictions.

Publishing the key alongside the analysis makes code-based linking straightforward and breaks this design. Even with a restricted key, unusual values or small groups may disclose identity through other information. No actual identity lookup, access-control enforcement or

deletion is performed here.

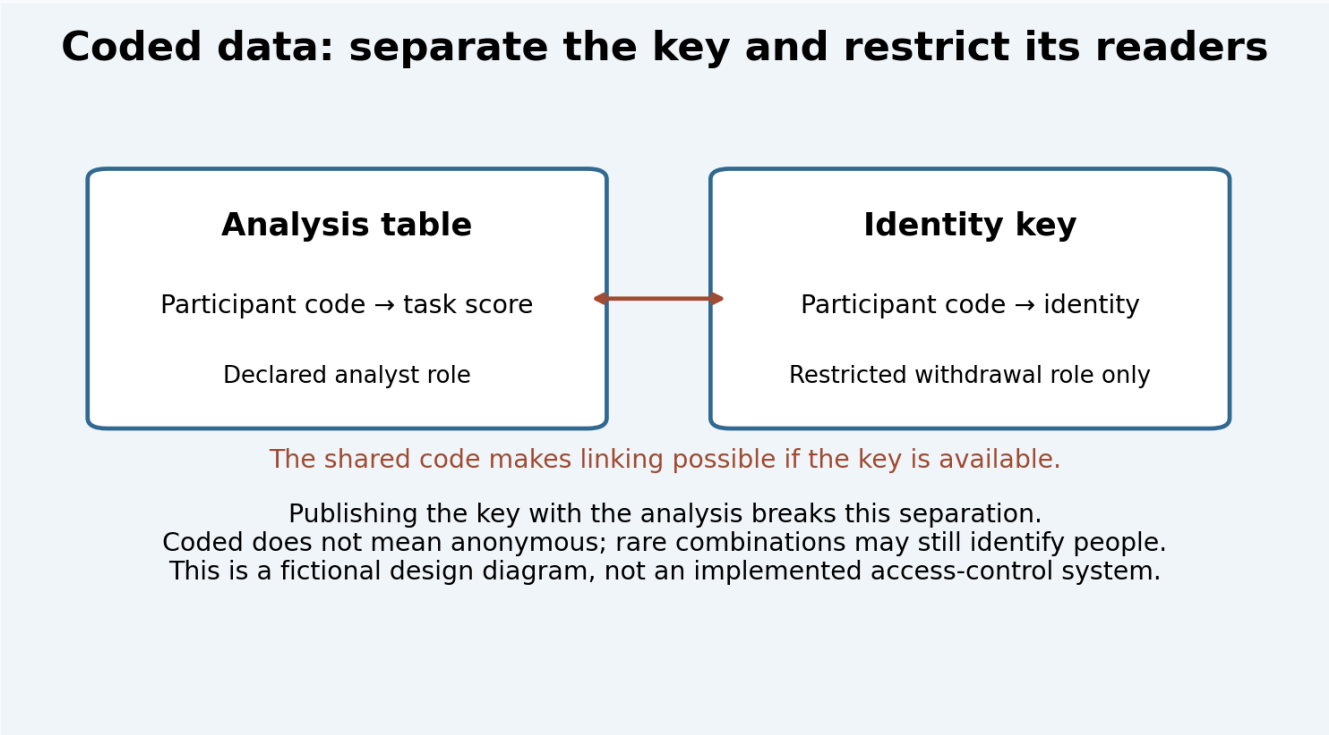
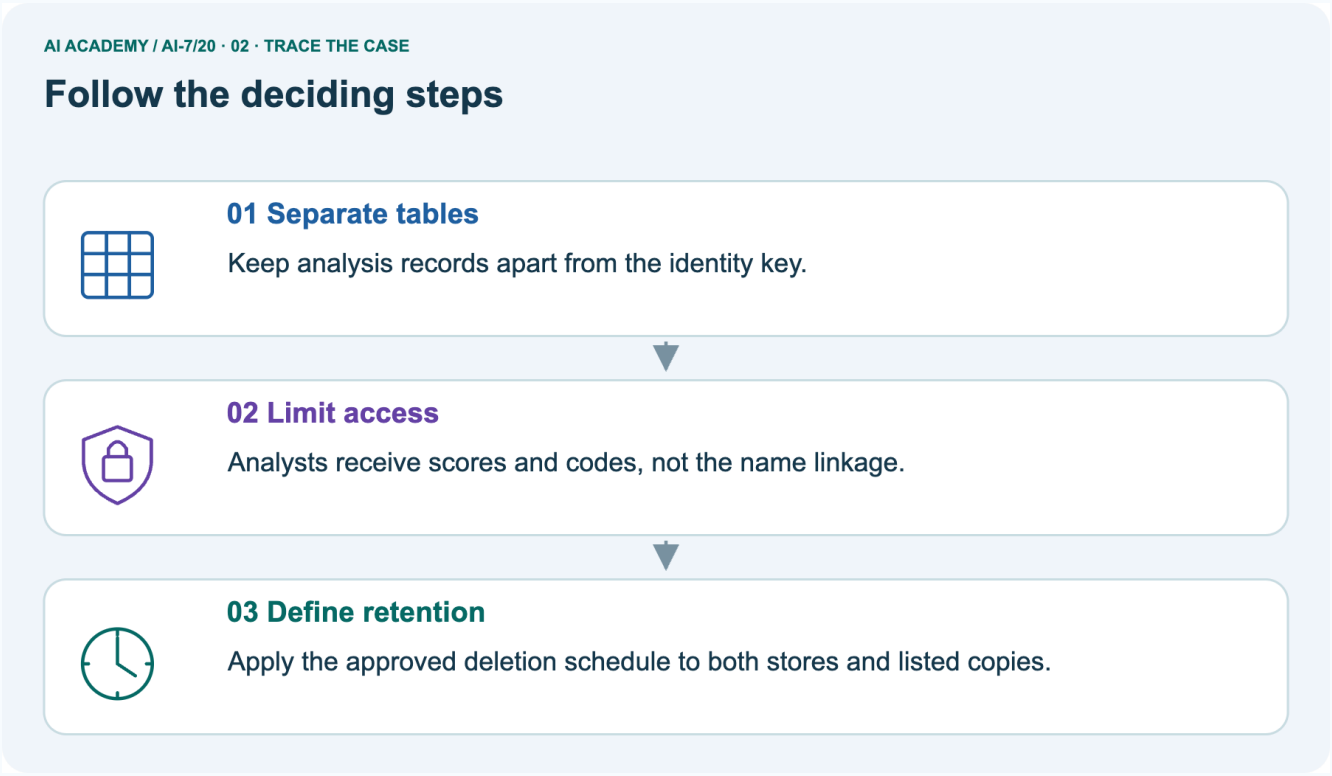


Table or operation	Declared permitted role/use	What the audit cannot assume
Analysis code + score	Analyst for the named task-score purpose	Codes are not proof of anonymity
Identity key	Restricted withdrawal role	A table label does not enforce permissions
Combine key and analysis publicly	Outside this design	Do not treat the result as deidentified
End-of-exercise deletion plan	Proposed retention action	Not evidence deletion has occurred
Subgroup count below five	Blocked by supplied policy	Changing the threshold after seeing outcomes is not compliance
Subgroup count equals five	Meets this size rule only	Not a general privacy or export guarantee

Picture trace and published-key explanation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate tables	Keep analysis records apart from the identity key.
2. Limit access	Analysts receive scores and codes, not the name linkage.
3. Define retention	Apply the approved deletion schedule to both stores and listed copies.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Change: Publish the identity key with the analysis table.

Prediction: The records become directly linkable to names and breach this access design.

Reason: Separation only helps while the linkage key remains restricted.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Data governance	Rules for purpose, access, quality and lifetime
Pseudonym	A code replacing direct identification
Disclosure risk	The possibility of learning identifying or sensitive facts

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

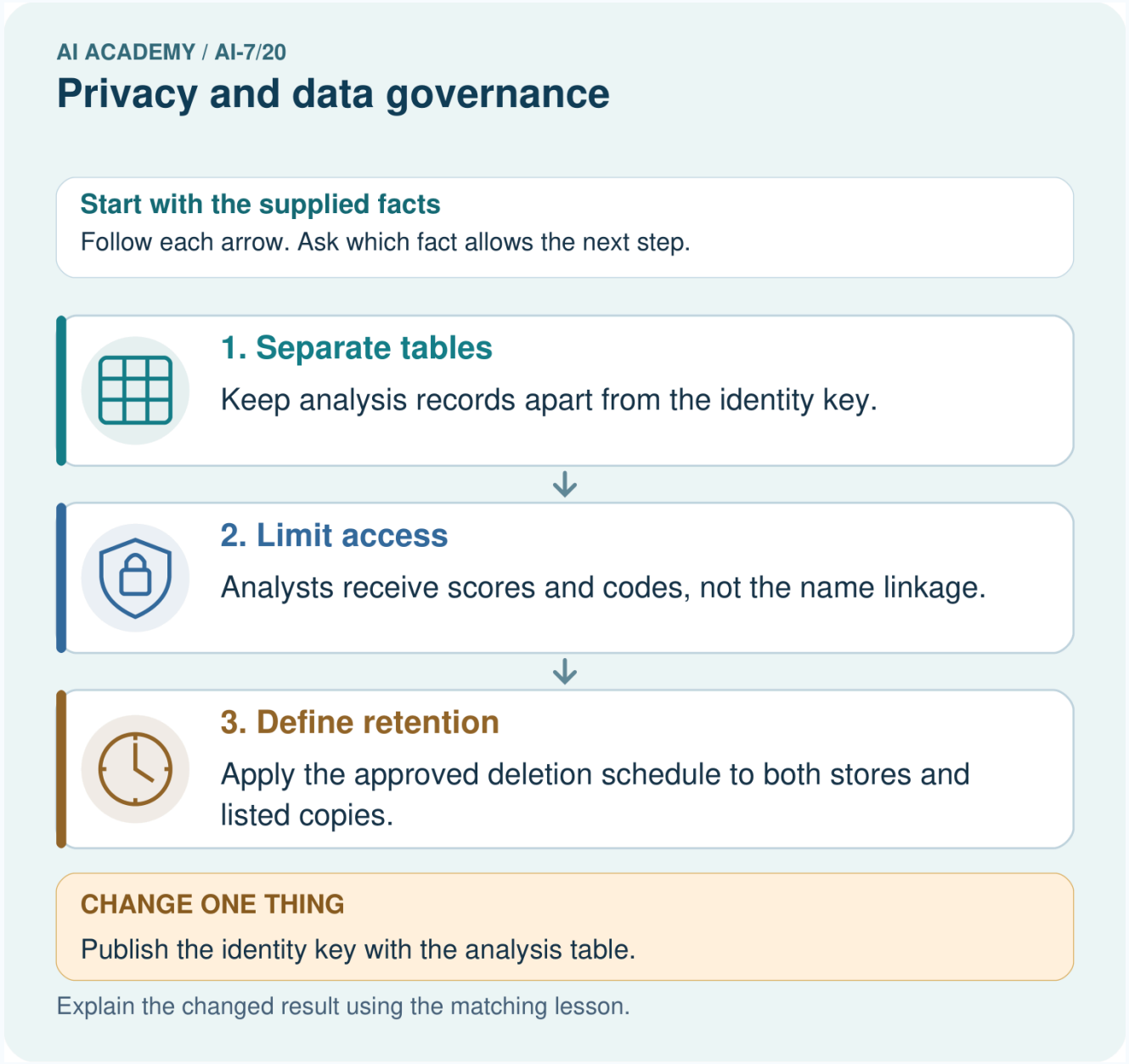
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Privacy and data governance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Privacy and **data governance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A size threshold or coding scheme alone does not guarantee privacy.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

Keep analysis records apart from the identity key. Analysts receive scores and codes, not the name linkage. Apply the approved deletion schedule to both stores and listed copies.

Changed case: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

The restricted linkage key makes these **pseudonym**ous records, not guaranteed anonymous ones.

Changed-case reasoning: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main coded combinations can remain identifying; name removal is insufficient. Picture key publication breaks the declared separation and enables linking. The executable audit checks metadata only. Independent one-member subgroup fails the minimum-five rule; do not report it.

A suppression or permitted regrouping proposal must consider indirect disclosure, and satisfying the size rule alone never establishes anonymity or export authorization.

Diagnose the current misconceptions

The worked result establishes every future case: A size threshold or coding scheme alone does not guarantee privacy.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

Teacher explanation: Do not release that subgroup table under the fictional policy. Combine only where meaningful and permitted, or omit with explanation.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Data governance** specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

W2 Trace the worked case

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

W3 Audit the limit

A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A size threshold or coding scheme alone does not guarantee privacy.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A size threshold or coding scheme alone does not guarantee privacy.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Qualification: A size threshold or coding scheme alone does not guarantee privacy.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

Teacher explanation: Do not release that subgroup table under the fictional policy. Combine only where meaningful and permitted, or omit with explanation.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Do not release that subgroup table under the fictional policy. Combine only where meaningful and permitted, or omit with explanation. Require intermediate reasoning and the stated scope.

Fresh retry: Subgroups sizes4 and6 under minimum5. Expected reasoning: Size4 fails the release policy; size6 satisfies that size rule but still needs other disclosure checks.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled privacy and **data governance evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of privacy and **data governance**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Predefining an analysis plan. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/21, ai-7/23, ai-7/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of privacy and **data governance**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Check every learner before discussion: Use the worked case for Privacy and **data governance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A size threshold or coding scheme alone does not guarantee privacy.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Publish the identity key with the analysis table. Require a **prediction** and a reason before comparing. Expected result: The records become directly linkable to names and breach this access design.

Reason: Separation only helps while the linkage key remains restricted.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

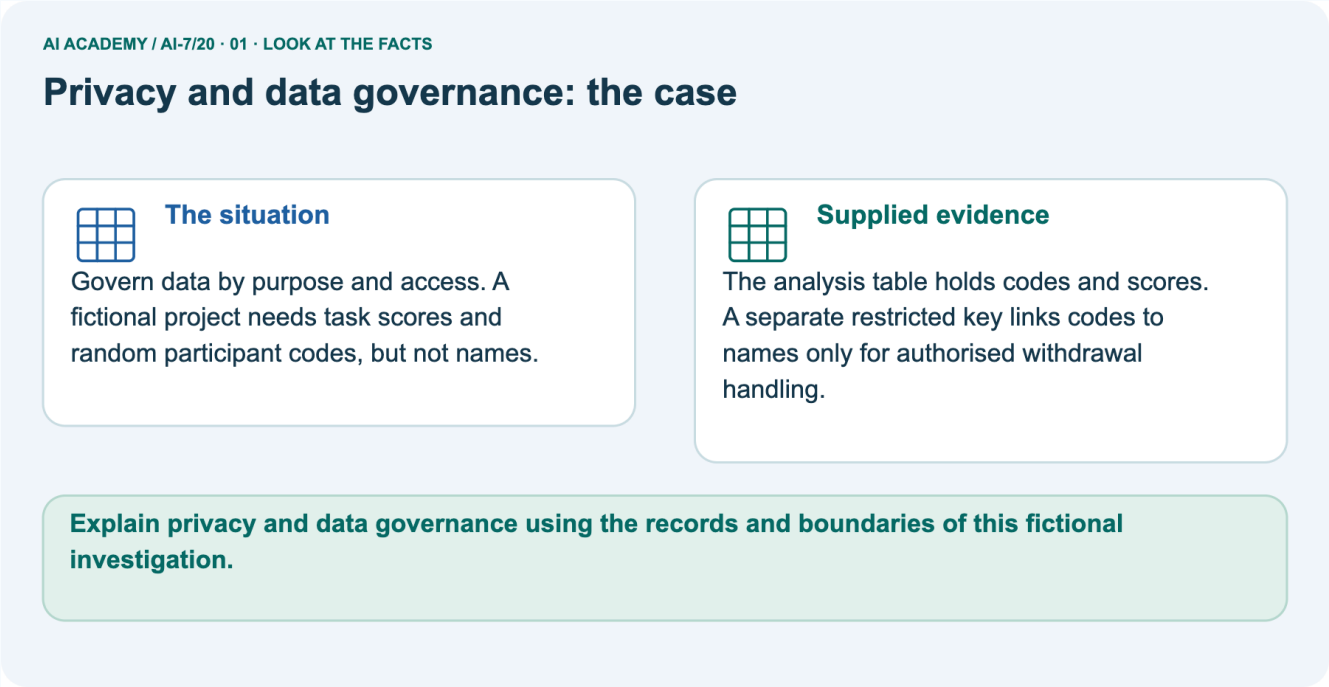
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Privacy and data governance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

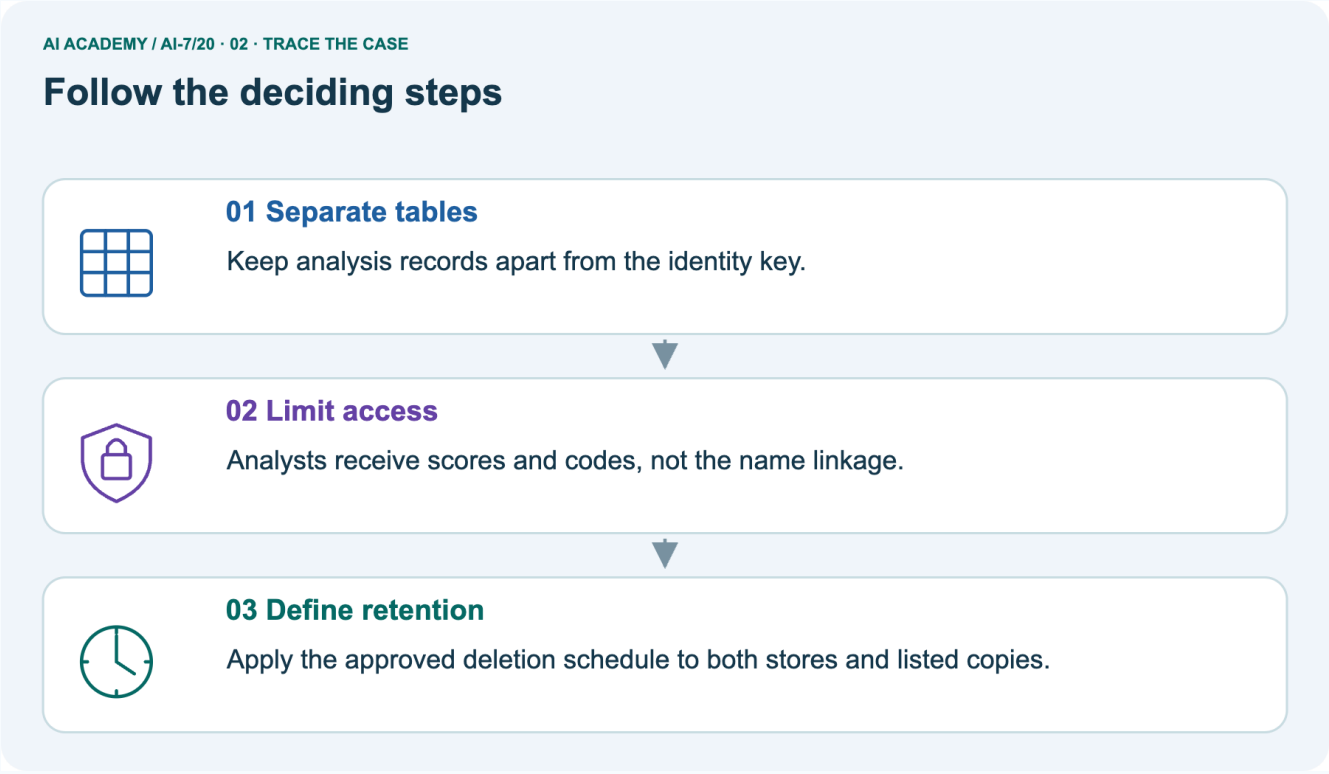
Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Supplied facts: The analysis table holds codes and scores.

A separate restricted key links codes to names only for authorised withdrawal handling.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



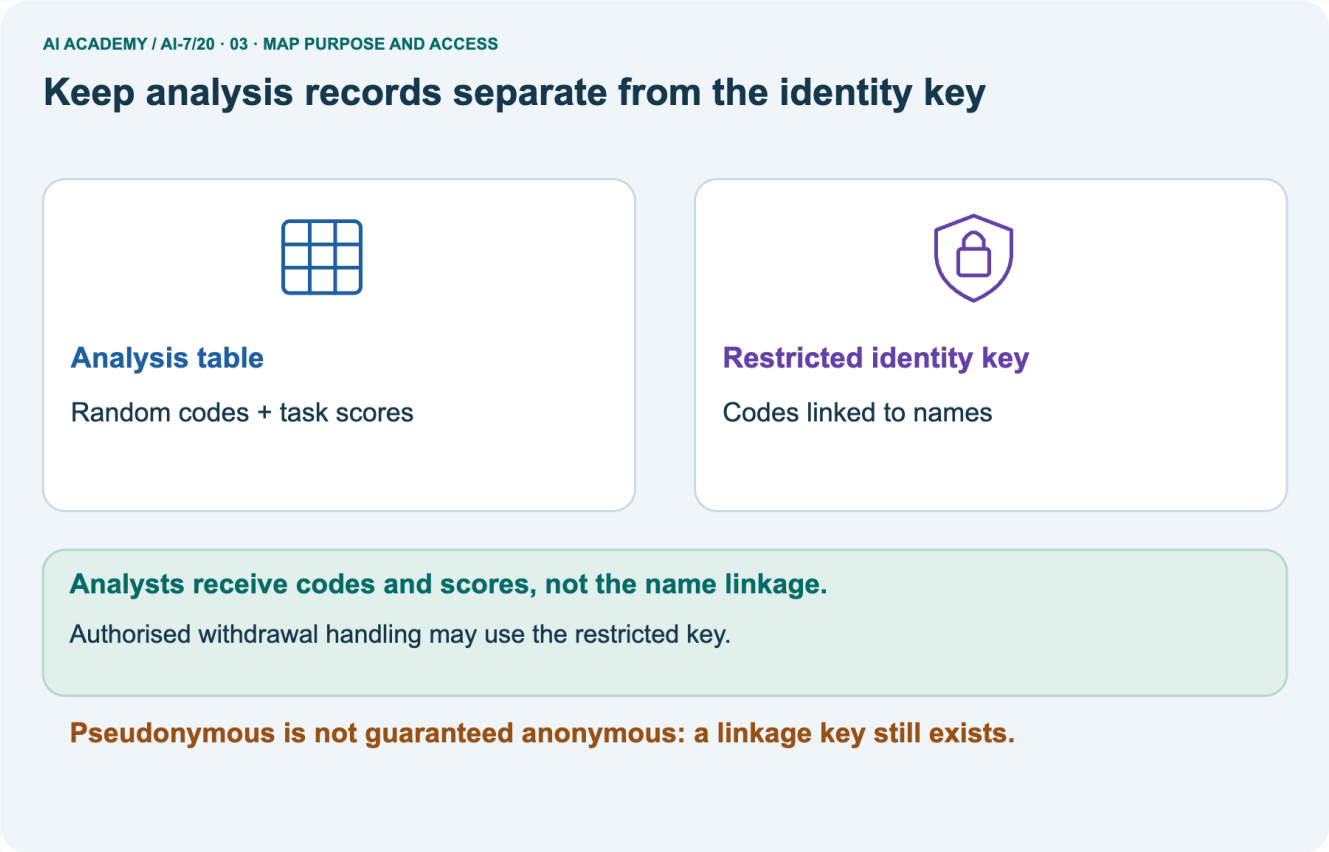
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate tables: Keep analysis records apart from the identity key. → Limit access: Analysts receive scores and codes, not the name linkage. → Define retention: Apply the approved deletion schedule to both stores and listed copies.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep analysis records separate from the identity key



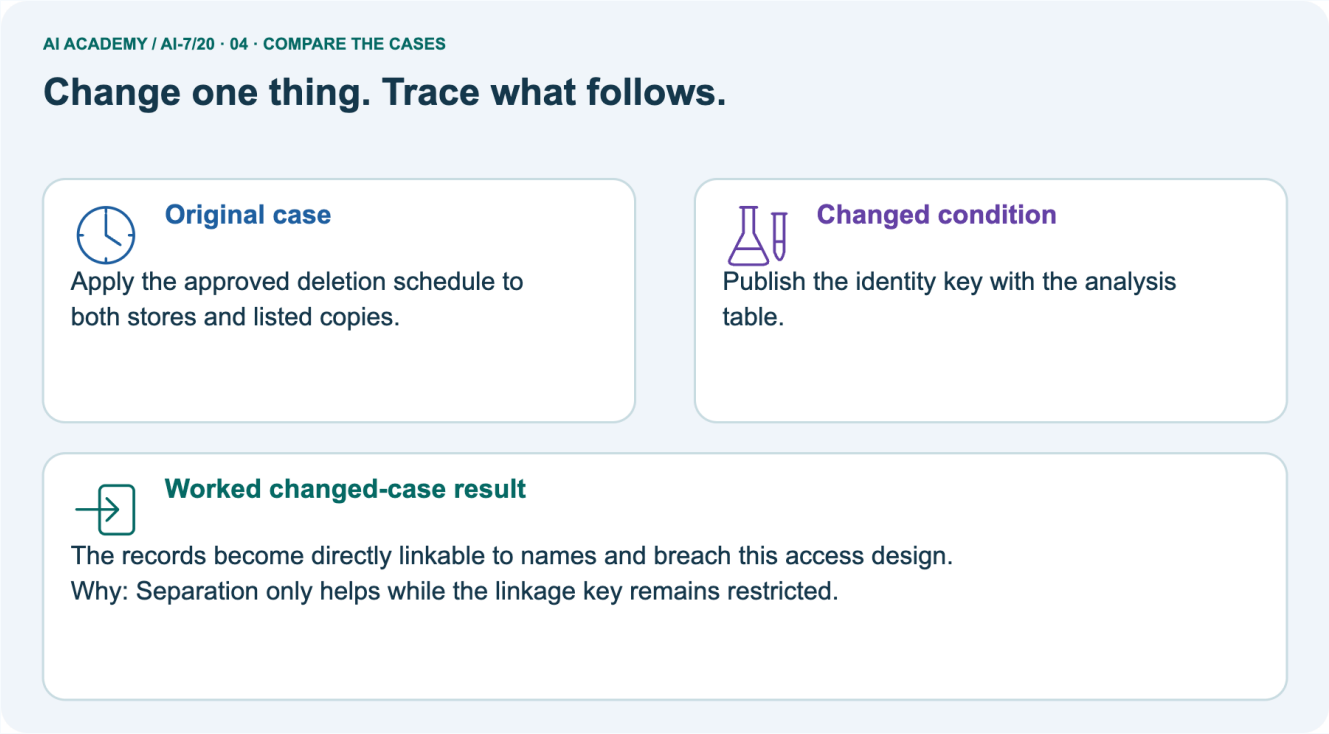
Follow the labelled picture with the matching explanation. Open the vector diagram

Analysts receive random participant codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling. Keeping the key restricted reduces exposure; it does not guarantee anonymity.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Apply the approved deletion schedule to both stores and listed copies. Changed condition: Publish the identity key with the analysis table.

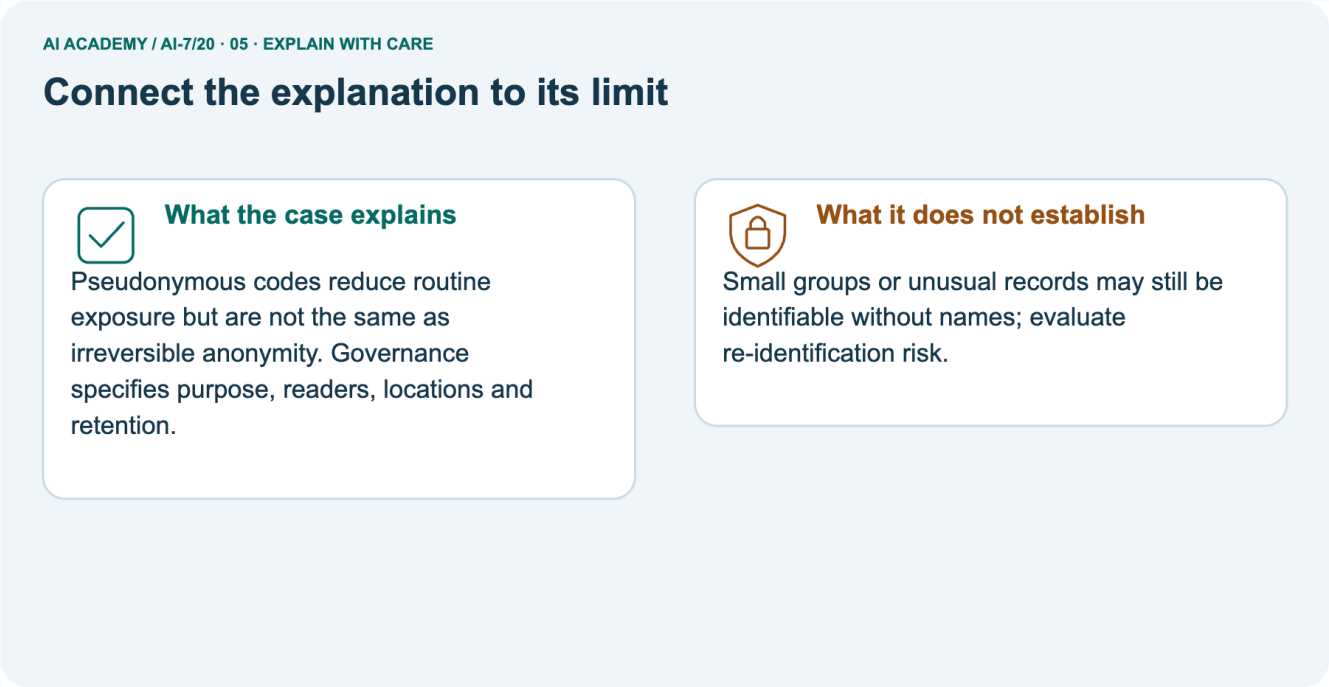
Expected result: The records become directly linkable to names and breach this access design. Reason: Separation only helps while the linkage key remains restricted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Separation only helps while the linkage key remains restricted.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

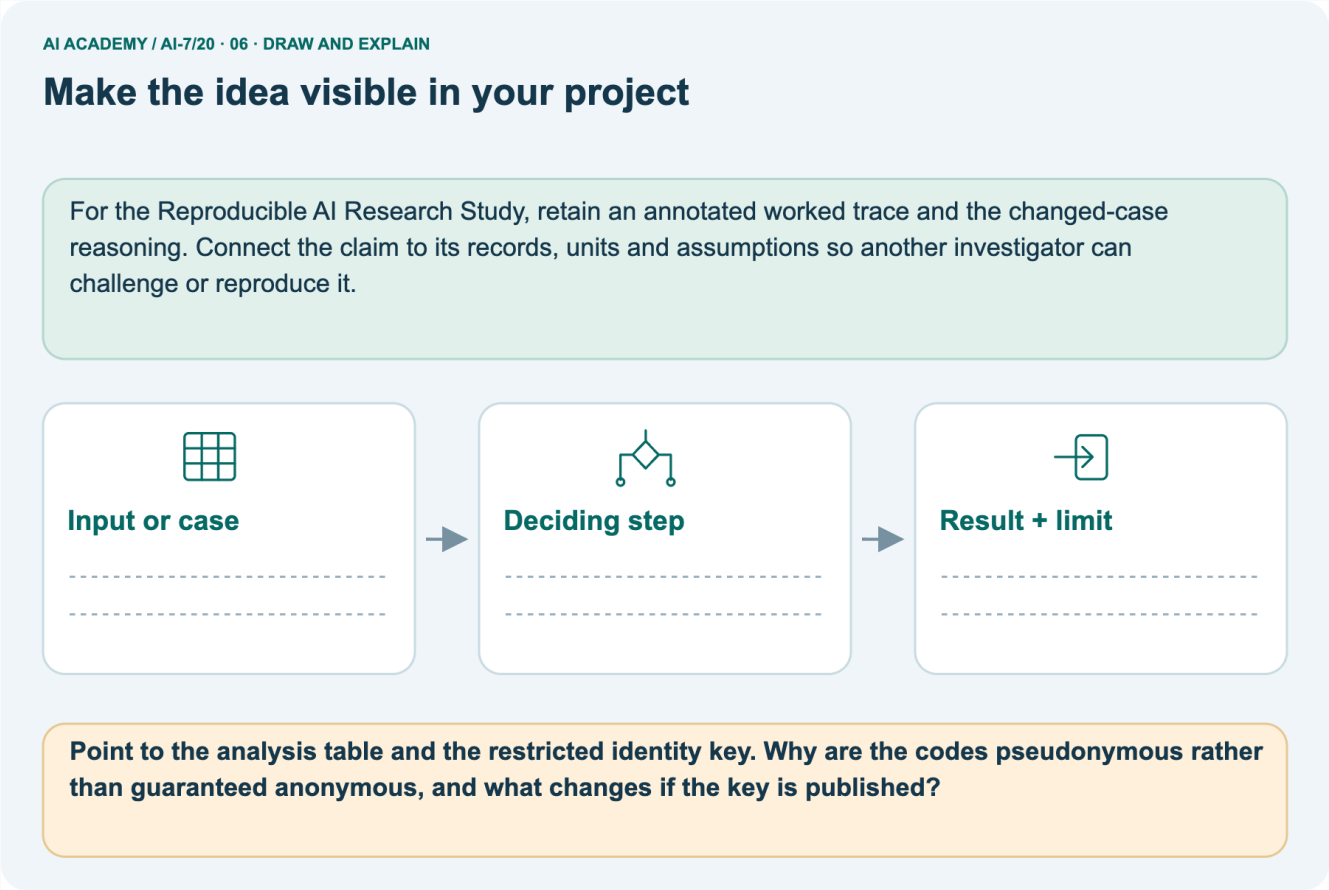
Limit: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the analysis table and the restricted identity key.

Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep analysis records apart from the identity key. Analysts receive scores and codes, not the name linkage. Apply the approved deletion schedule to both stores and listed copies.

Changed case: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

The restricted linkage key makes these **pseudonym**ous records, not guaranteed anonymous ones.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Privacy and data governance: the case

02. Follow the deciding steps

03. Keep analysis records separate from the identity key

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.